

Documentation for Themed UI Components: Digital Clock and Factorial Calculator

1. Introduction

The Digital Clock and Factorial Calculator are two interactive UI components integrated with a customizable theme system. The user can toggle between various themes to alter the look and feel of the application. Themes include Dark, Light, Retro, Ocean, Sunset, and Cyberpunk. The theme not only changes the background but also modifies UI elements such as the digital clock display, input fields, buttons, and results.

2. Features

- Digital Clock: Displays the current time, formatted in hours, minutes, and seconds. It dynamically adjusts based on the selected theme to display glowing or shadowed effects for an engaging user experience.
- Factorial Calculator: A simple calculator where users can input a number, and the factorial of that number is displayed. Input fields, buttons, and the result area change dynamically according to the selected theme.

3. Themes

The following themes are available for the components:

Dark Theme (bg-dark)

Background: Dark gray

Text: White

Components: Input fields and buttons have dark, muted colors with bright text for visibility.

Light Theme (bg-light)

Background: Light gray or white

Text: Dark gray or black

Components: Light backgrounds with subtle borders for inputs and buttons.

Retro Theme (bg-retro)

Background: Black with a vintage glow effect

Text: Red with shadow effects

Components: Retro-style input and button elements with bright neon-like effects.

Ocean Theme (bg-ocean)

Background: Deep ocean blue

Text: Light blue or white

Components: Ocean-inspired blue tones with bright buttons and input fields.

Sunset Theme (bg-sunset)

Background: Warm sunset orange

Text: Peach or light colors

Components: Warm, glowing text and interactive elements styled with vibrant sunset hues.

Cyberpunk Theme (bg-cyberpunk)

Background: Dark blue with a neon pink highlight

Text: Neon pink

Components: Bold, cyberpunk-inspired visuals with glowing neon pink accents.

4. CSS Styling

The styling is applied through a set of custom CSS classes. The CSS is organized to target both global body elements and specific components like the clock, input fields, buttons, and result sections.

5. JavaScript Functionality

The theme can be toggled through JavaScript by switching the `body` class between theme classes (e.g., `bg-dark`, `bg-light`, etc.).

Theme Switching:

```
function changeTheme(theme) {  
  document.body.className = theme;  
}  
function updateClock() {  
  const clock = document.getElementById('clock');  
  const time = new Date().toLocaleTimeString();  
  clock.textContent = time;  
}  
setInterval(updateClock, 1000);
```

Factorial Calculator:

```
function calculateFactorial() {  
  const number = document.getElementById("inputNumber").value;  
  let result = 1;  
  if (number === "" || number < 0) {
```

```

    document.getElementById('result').innerText = 'Please enter a valid number';
    return;
}
for (let i = 1; i <= number; i++) {
    result *= i;
}
document.getElementById('result').innerText = `Factorial of ${number} is: ${result}`;
}

```

6. HTML Structure

Digital Clock HTML:

```
<div id="clock"></div>
```

Factorial Calculator HTML:

```

<div id="factorial-calculator">
  <h3>Factorial Calculator</h3>
  <input type="number" id="inputNumber" placeholder="Enter a number" />
  <button onclick="calculateFactorial()">Calculate Factorial</button>
  <div id="result"></div>
</div>

```

7. How to Use

1. Theme Switching: The theme can be changed by calling the `changeTheme(theme)` function with one of the available theme names (e.g., 'bg-dark', 'bg-light').
2. Digital Clock: The clock will automatically update every second. It reflects the currently selected theme by adjusting the text-shadow for glowing effects.
3. Factorial Calculator: Enter a number in the input field and click Calculate Factorial. The result will be displayed below the button, and the UI elements will adjust according to the active theme.

8. Customization

To add new themes or adjust existing ones:

1. Modify CSS: Add new CSS classes with the desired colors, text effects, and other styles.
2. Update Theme Switcher: Ensure the new theme is included in the `changeTheme` function.
3. Adjust Components: If necessary, adjust the styles of the clock or factorial calculator components to reflect any changes made to the theme.

9. Conclusion

This implementation provides an interactive and visually dynamic user interface for both a Digital Clock and a Factorial Calculator. With the use of custom themes, users can switch between different styles to enhance their user experience. The solution is highly customizable and provides a professional, user-friendly interface with real-time theme switching, seamless transitions, and responsive components.

Contact Information

For further inquiries or support, feel free to reach out via the official support channels.